Capstone 6 – Verification Report

1. Question

Can AI explain beginner Python code clearly and correctly, and where does it mislead learners?

2. Method

1. Ten short programs (loops, functions, lists, dictionaries, file handling, strings) were selected from the style of public tutorials.
2. Each program was run in Python 3 (Google Colab-compatible) to capture the **real** output.
3. Fifteen statements about program output were compared with the real output using `verify_claim()` (see document 11).
4. Every program and several sample snippets were scanned with `scan_risks()` for outdated syntax and unsafe patterns.
5. Practice problems were checked automatically with `check_answer()`.

3. Results

Measure	Result
Programs that ran without error	10 / 10
Output claims checked	15
Correct claims	10 (66.7%)
Wrong claims	5 (33.3%)
Risk-scanner detections on sample snippets	5 / 5
Course programs flagged by scanner	1 (P09, mutable default)
Faulty practice solution caught by checker	Yes (1/4 tests passed)

4. Where explanations mislead

Category	Programs	Why it misleads
Boundary errors	P01, P04	End values excluded in `range` and slices; items shift after `insert`
Special cases	P03	0 does not skip the function; loop is empty and result stays 1
Modes and hidden state	P08, P09	`"a"` vs `"w"`; shared default list
Outdated syntax	scanner samples	Python 2 `print`, `raw_input`, `xrange`
Unsafe suggestions	scanner samples	`eval(input())`, `os.system(...)`

5. Findings

- Simple linear programs (P02, P05, P06, P07, P10) were easy to explain correctly.
- Errors appear where behaviour is **implicit**: excluded end values, shared defaults, file modes.
- Unsafe or outdated code is easy to spot automatically but easy to miss by reading.
- Running the code and comparing outputs found every wrong claim in the test set.

6. Limits of this report (be honest in your presentation)

- The 15 claims were written to represent typical correct and incorrect statements; **they are not a measured error rate of any specific AI tool.**
- To measure a tool, paste its real answers into `verify_claim()` and record the verdicts. Compare at least 4 tools (ChatGPT, Gemini, Claude, plus a Colab-based check) in the table below.
- Ten short programs is a small sample.

7. Tool comparison (fill with your real observations)

Tool	Programs tested	Output claims correct	Wrong	Outdated/unsafe seen
ChatGPT (free)				
Gemini (free)				
Claude (free)				
Google Colab (ground truth)	10	10 real runs	0	n/a

8. Conclusion

AI is a fast first explainer but not a trustworthy final authority. Beginners should treat each explanation as a hypothesis, run the code, and use the checker. Responsible use means disclosing which parts the AI wrote and which parts were verified.